**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Wired and Wireless Networks

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

1. What happens if the central switch of a star network fails?

2. More users share a limited network link. What may happen?
   A) More bandwidth per user
   B) Less bandwidth per user
   C) Each device gains RAM
   D) All delays disappear

3. What distinguishes a WAN?

4. Convert denary 150 into eight-bit unsigned binary.

## Today's objectives

Distinguish copper, fibre-optic and wireless connections. Explain suitable advantages and drawbacks for each.

## Key vocabulary - Read, cover, recall

**Wired connection**
A link using a physical cable.

**Copper cable**
Transmission medium carrying electrical signals.

**Fibre optic**
Transmission medium carrying signals as light.

**Wireless connection**
Communication without a connecting cable, often radio.

**BLACK PEN: Write one definition in your own words.**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New Learning - Key ideas

**Copper cables**

Wired copper connections carry electrical signals and are affordable, but have speed/distance limits.

**Fibre-optic cables**

Fibre carries pulses of light, supporting high rates and long distances; installation can be more costly.

**Wireless connections**

Wi-Fi uses radio waves so devices can move without cables, but connections may experience interference and interception.

**Choose a connection**

Use the scenario: high bandwidth, cost, mobility, length of cable, interference and security matter.

## I Do - Worked model

**A college connects two buildings a long distance apart. Explain why fibre optic may be more suitable than copper. [4 marks]**

- The connection is between buildings over a substantial distance.
- Fibre uses light pulses [1] and is suitable for high-speed, longer-distance links [1].
- Copper carries electrical signals [1] and has greater limitations on speed/distance for such a link [1].
- Two contrasts, each explained in context, make four points.

## We Do - Guided practice

1. A cafe provides wireless access to customers using phones and tablets. Explain two reasons Wi-Fi is suitable. [4 marks]

2. A business handles confidential data over Wi-Fi. Explain two risks and how encryption helps. [4 marks]

3. A hospital must connect fixed machines reliably. Explain one benefit and one drawback of a wired connection. [4 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Predict the ideal transfer time in seconds.**

```
fileMb = 120
speedMbps = 60
print(fileMb DIV speedMbps)
```

## You Do - Independent exam questions

Q2. State one disadvantage of wireless transmission. [1 mark]

Q3. A research station must connect an outdoor field laboratory to its main building. Explain one advantage and one limitation of using fibre-optic cable for this long link. [4 marks]

Q4. Compare copper, fibre and wireless communication for a school. [6 marks]

## Exit Ticket - True or False?

A wireless connection always needs a separate cable from every phone to the access point.

**TRUE [ ]** **FALSE [ ]**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

## Do Now - Model answers

1. If the central switch fails, devices using it lose contact. A broken cable usually affects just one workstation.

2. B - Less bandwidth per user

3. A WAN links networks over a large area, often using links provided by other organisations.

4. (a) 10010110.

## I Do - Explain the model

The connection is between buildings over a substantial distance.

Fibre uses light pulses [1] and is suitable for high-speed, longer-distance links [1].

Copper carries electrical signals [1] and has greater limitations on speed/distance for such a link [1].

Two contrasts, each explained in context, make four points.

## We Do - Model answers

1. Wireless uses radio rather than cables to each device [1], allowing users to move while connected [1]. Devices can join without dedicated physical ports [1], making guest access more flexible [1].

2. Wireless signals can be intercepted by nearby devices [1], creating a confidentiality risk [1]. Encryption scrambles transmitted data [1], so captured content cannot be understood without the key [1].

3. Wired cables provide a stable link less affected by radio interference [1], improving reliability [1]. The cable restricts device movement [1], reducing flexibility if machines are relocated [1].

## Code First - Answer

fileMb = 120
speedMbps = 60
print(fileMb DIV speedMbps)
Expected output: 2

## You Do - Self-mark

Q2: Interference, interception/security risk or variable signal quality [1].

Q3: Fibre carries light-based signals [1] and is well suited to high-speed/long-distance links [1]. Installing fibre can be more costly or difficult [1], increasing the cost/complexity of the research station's connection [1].

Q4: Copper carries electrical signals [1] and is cost-effective for local wired links [1]. Fibre carries light [1] and offers high speed/long distance [1]. Wireless uses radio [1] and offers mobility but may suffer interference/interception [1].

**GREEN PEN: Choose the hardest answer. Correct it on lined paper.**